Indeed, a fundamental problem of applying human-centric tests and concepts to LLMs—such as theory of mind or emotional understanding—is the assumption that they engage with information in ways similar to humans, presupposing background psychological mechanisms that may be absent or irrelevant for LLMs (**Box 1**). These anthropomorphic assumptions undermine the construct validity (i.e., the degree to which a test measures the specific psychological construct it purports to measure) of psychological tests in LLMs (Millière & Buckner, 2024b). For example, response confidence in LLMs, as measured by token probability, may differ from human self-reports. Similarly, emotional intelligence—the ability to perceive, understand, manage, and use emotions in oneself and others—encompasses self-awareness, empathy, emotional regulation, and social skills, all rooted in subjective experiences that are absent in LLMs. Model "understanding" of emotions is purely expressive and syntactic, driven by learned associations between words and concepts rather than experiential, visceral insight or an internalized understanding of mental states. So, when LLMs perform like humans on tests of emotional understanding—in either overall score or response pattern (Wang et al., 2023)—this alignment is at the surface level, rather than reflecting genuine equivalence.

Apparent alignments between LLM and human responses may also be artifacts of specific task formulations rather than evidence of robust, human-like reasoning capabilities. One clue for a lack of true competence is a dissociation between accuracy and the model's explanations of its responses—namely, reasoning (Leivada et al., 2023). Another critical test is prompt sensitivity, examining how performance is affected by superficial alterations to the prompts—changes to which humans typically show little to no sensitivity or respond to differently. Such prompt sensitivity has been documented in LLMs in survey responses—RLHF-tuned models can be highly sensitive to changes like typos (Tjuatja et al., 2024)—and in tasks like reasoning (Binz & Schulz, 2023a; Dasgupta et al., 2022; Yax et al., 2024), decision making (Binz & Schulz, 2023b; Suri et al., 2024), theory of mind (Strachan et al., 2024), moral judgments (Oh & Demberg, 2025), and more (Kamoi et al., 2024; McCoy et al., 2024).

For example, LLMs but not humans improved their performance in reasoning tasks when the instruction included the phrase "let's think step by step" (Yax et al., 2024), a form of chain-of-thought prompting. Conversely, while LLMs performed at ceiling in a false belief test just like human participants, they struggled when small changes were made to the formulation of false belief scenarios—suggesting syntactic pattern processing rather than robust reasoning (Strachan et al., 2024). These sensitivities to minor variations highlight how apparent alignment can mask differences in underlying processes, creating a threat to internal validity—the ability to draw firm conclusions about causal relationships—when the observed performance is misattributed to the intended manipulation rather than to superficial linguistic patterns.

But even when performance reflects underlying abilities, it is not clear whether models employ mechanisms similar to those of humans. Given that different systems can achieve the same outcome through different mechanisms—known as multiple realizability (Bowers et al., 2023; Guest & Martin, 2023), as in telling time with digital versus mechanical clocks—it is unwarranted to assume mechanistic or even functional equivalence (**Box 4**). Indeed, it is notoriously difficult to understand exactly what LLMs have learned. Beyond inherent differences between machine and biological intelligence in their architecture and algorithms, LLMs are trained on datasets much larger than what human learners experience. Differences in mechanisms may manifest as distinct response characteristics, including (1) context sensitivity, such as prompt sensitivity or performance variations across different vignettes (Yax et al., 2024); (2) response patterns, such as variability in open-ended responses (Li et al., 2024), item-by-item performance variability (Wang et al., 2023), and correlation of accuracy with confidence (Yax et al., 2024); and (3) error types and consistency, such as errors arising from cognitive demands versus those from item wording or familiarity (Yax et al., 2024).

This raises a fundamental question: When LLMs perform at human levels in psychological tasks, what does it tell us about the capacity of these models? Performance can reflect true competence—some underlying abilities—or something more superficial, like pattern memorization (Gao et al., 2024) or other surface-level cues or pure chance. Conversely, underperformance can reflect something other than incompetence, such as processing limitations or ineffective prompting (Firestone, 2020). Thus, to establish true competence, model outputs should be sensitive to changes in the task-relevant inputs but insensitive to irrelevant changes (Harding & Sharadin, in press).

To mitigate the alignment-as-explanation fallacy, safeguards include testing alignment robustness against perturbations, examining reasoning processes, using causal interventions, performing cross-domain validation, and explicitly acknowledging the limits of inferring shared mechanisms. Specifically:

1. **Model testing and validation**: Test whether the alignment between LLM and human responses is robust to perturbations in task structure, prompt wording, and contextual variations that shouldn't affect performance (Oh & Demberg, 2025). If the performance varies significantly with superficial changes, this suggests a lack of true construct equivalence (Firestone, 2020).
2. **Process tracing**: Use methods such as chain-of-thought prompting to examine the reasoning paths that LLMs use to arrive at answers, comparing these with human reasoning protocols (Bao et al., 2024). Significant differences in reasoning processes, even when outputs align, would suggest different underlying mechanisms.
3. **Causal interventions**: Implement interventions that test specific hypotheses about psychological mechanisms. For example, if humans and LLMs appear to use similar heuristics, test whether manipulations designed to affect those heuristics produce comparable effects (GX-Chen et al., 2025). Examine the causal effects of training data on model output through fine-tuning (Hu et al., 2025).
4. **Cross-domain validation**: Examine whether alignment in one domain generalizes to related domains that should engage similar cognitive processes. Limited transfer may suggest different underlying mechanisms despite surface alignment (Zan et al., 2025). The same applies to validation across different languages of the prompt, such as English and Chinese (Jin et al., 2024).
5. **Explicit limitation acknowledgment**: When reporting alignments between LLM and human responses, acknowledge the limitations of inferring shared mechanisms, noting their distinct architecture and learning history (Abdurahman et al., 2025).

Box 4 | Functional versus Mechanistic Equivalence

Functional equivalence occurs when two systems produce the same outputs given the same inputs under specified conditions—like how both a digital watch and a sundial tell accurate time on a clear day, despite different mechanisms. Mechanistic equivalence, in contrast, requires shared internal causal processes—addressing *how* results are produced, not merely *that* they are produced. Because distinct mechanisms can yield identical functions (multiple realizability), functional alignment alone cannot establish mechanistic alignment.

In psychological research, an LLM that correctly answers false belief questions demonstrates functional alignment with humans, but evidence from prompt sensitivity studies reveals mechanistic divergence. Even "perfect" behavioral correlation therefore licenses, at most, a claim of functional parity. Demonstrating mechanistic parity would demand convergent evidence from: (1) input perturbation tests—performance should degrade in the same pattern as humans when task-relevant cues are modified; (2) internal-state homology—representational similarity analyses should reveal shared information coding; and (3) causal intervention—disabling a model component should mirror the effect of corresponding cognitive impairments in humans (Firestone, 2020; Guest & Martin, 2023).

Differentiating functional and mechanistic equivalence has implications for three levels of LLM–human similarity:

- **Level 1: Pragmatic simulation.** If the research goal is purely functional—pre-testing vignettes, forecasting average responses, relationship counseling—then output equivalence between LLMs and humans may suffice regardless of underlying mechanisms. Here, the simulation perspective is interested primarily in *what* is produced, not *how* it is produced. For example, if an LLM reliably predicts which survey items show ceiling effects, its utility as a rapid prototyping tool remains valid even if its internal processes differ from human cognition.
- **Level 2: Explanatory inference.** If the research goal is theoretical—testing models of cognition, emotion, or moral reasoning—then superficial behavioral alignment becomes insufficient. Internal and construct validity demand evidence that similar underlying mechanisms are at play. An LLM might generate human-like responses to reasoning tasks without engaging in human-like reasoning—just as a calculator produces accurate math answers through processes entirely unlike human calculation.
- **Level 3: Phenomenological attribution.** Even if perfect functional and mechanistic overlap were demonstrated, this would not automatically confer phenomenal states—subjective experiences like visceral, emotional empathy. Claims that LLMs “have beliefs,” “feel empathy,” or “experience understanding” involve a third level of equivalence that requires additional, presently unknown properties (e.g., global workspace dynamics, biological embodiment, or other factors that might give rise to consciousness). Consequently, attributions of subjective experience to LLMs remain largely speculative until such criteria are articulated and empirically evaluated.

Thus, LLMs can serve as valuable functional simulators for many research purposes without needing to process and experience the world as humans do. However, researchers must be transparent about which level of equivalence they are claiming and provide appropriate evidence for that level. The danger lies not in using LLMs as Level 1 simulators, but in making unwarranted leaps to Level 2 theoretical inferences or Level 3 phenomenological attributions—shortcuts that threaten validity.

Anthropomorphism Fallacy

Beyond misalignments due to algorithms, purposes, and implementations, another issue with the replacement view is that it leads to anthropomorphism of LLMs (Crockett & Messeri, 2023; Shiffrin & Mitchell, 2023). Their fluent, human-seeming responses can trigger an enhanced ELIZA effect: the tendency to attribute human-like understanding to systems that merely manipulate symbols, creating a compelling “language user” illusion (**Box 1**). Indeed, when researchers refer to “the minds of language models” or “the machine minds of LLMs” (Dillion et al., 2023), such terminology—whether used metaphorically or literally—can inadvertently encourage teleological bias, attributing purposes and goals to LLMs.